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Audirac

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(54) **DEVICE FOR LOCKING A DOOR**

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(52) **U.S. Cl.**

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(2013.01); **E05B 9/002** (2013.01); **E05B 17/06**
(2013.01);

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1/02; E05C 1/06; E05C 1/08;

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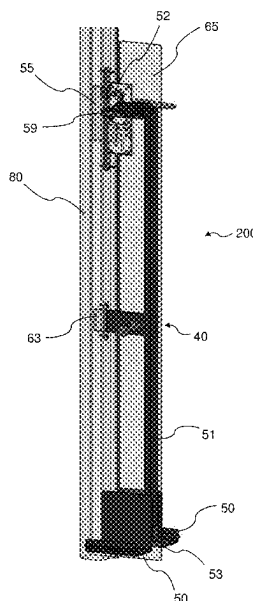
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Pinchas

(57)

ABSTRACT

The invention relates to a device (200) for locking a door, which comprises: a bolt (40) comprising a bar (52) configured to engage in a housing secured to the frame (80) accommodating the door and a pusher means secured to the bar (41) and spaced apart from the bar by a stem and in which the bolt is positioned on the door at a height relative to the foot of the door of between thirty and eighty centimetres, such that the bar of the bolt can be engaged with or disengaged from the housing secured to the frame by pressure applied by a user's knee against the pusher means.

14 Claims, 11 Drawing Sheets



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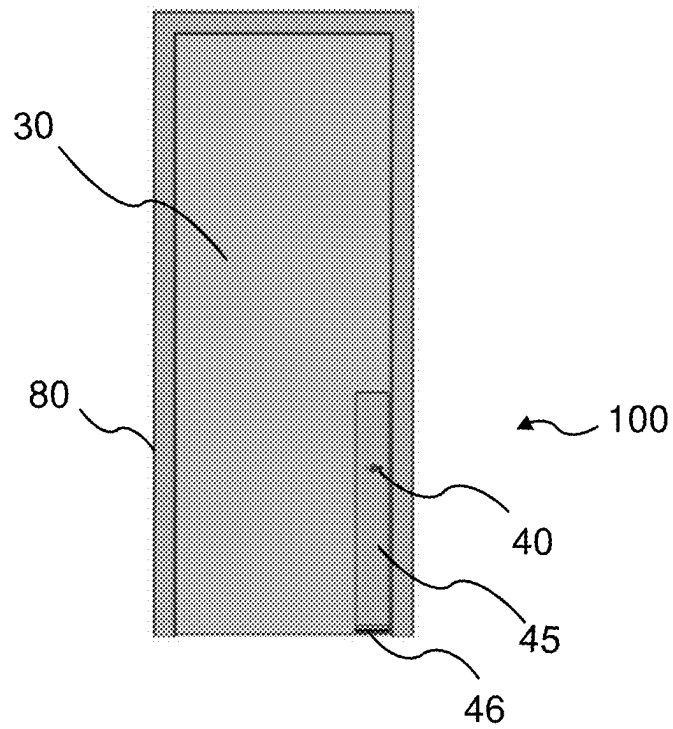


Fig. 1

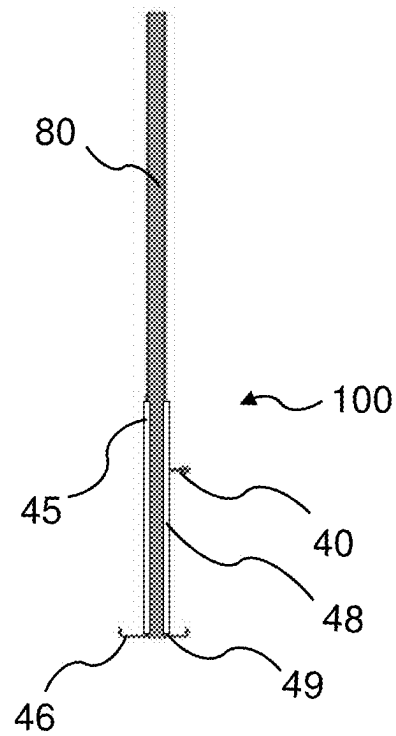


Fig. 2

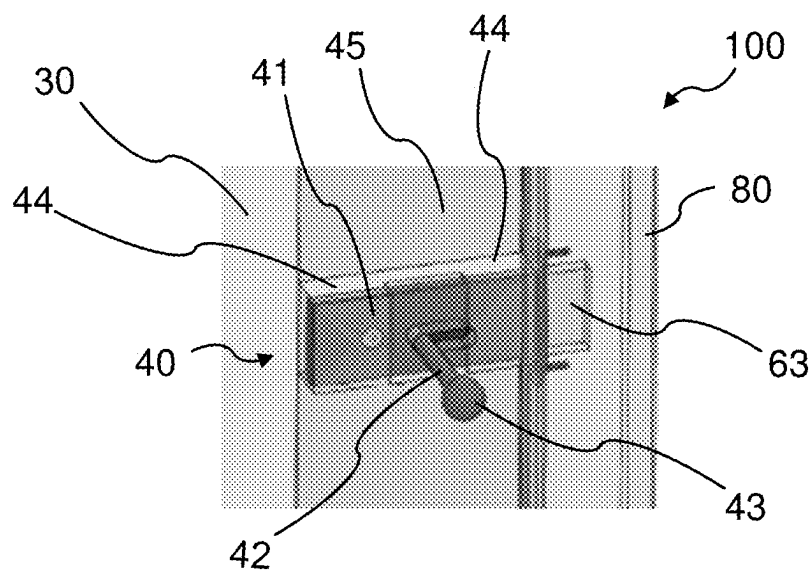


Fig. 3

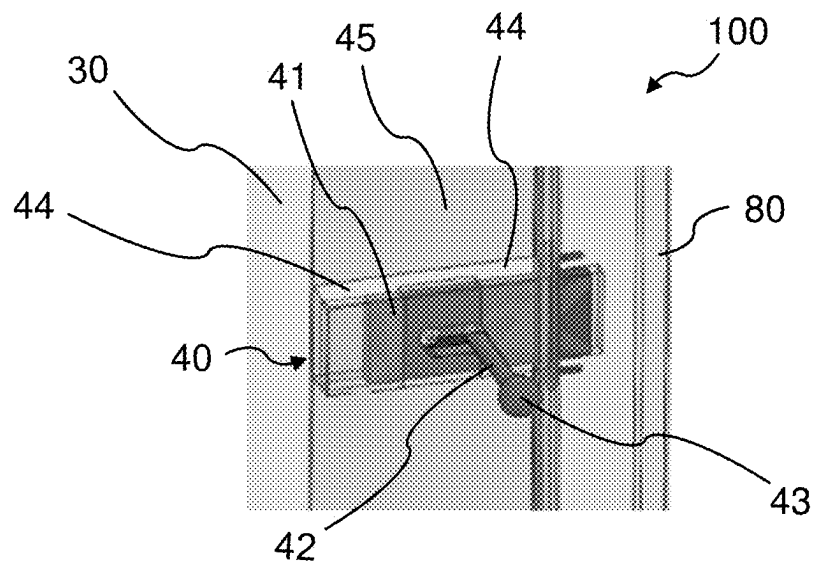


Fig. 4

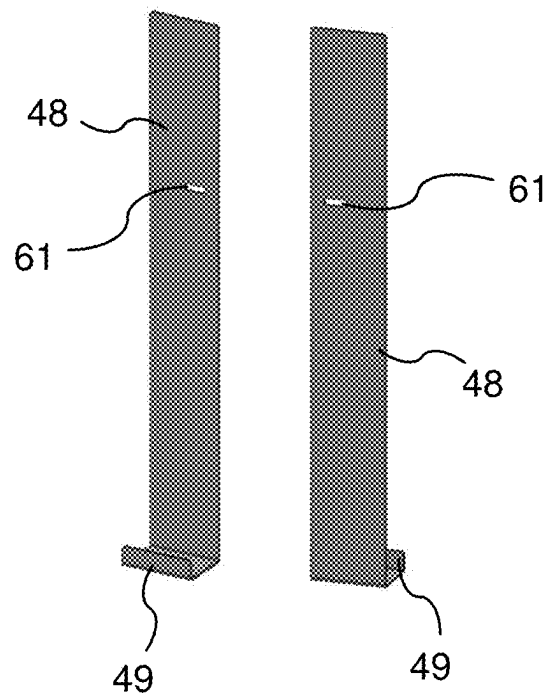


Fig. 5

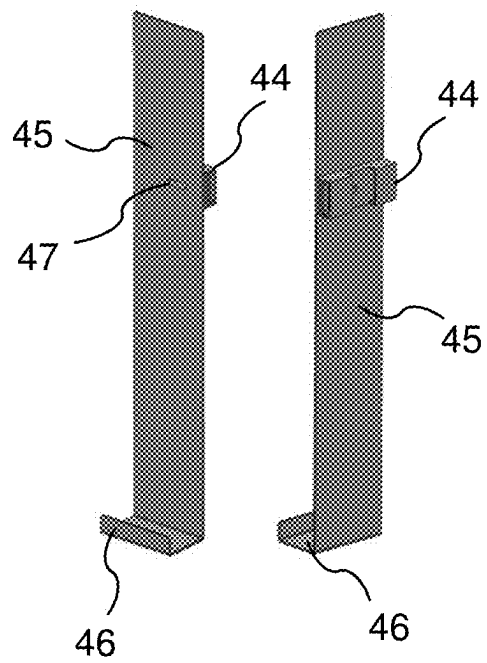


Fig. 6

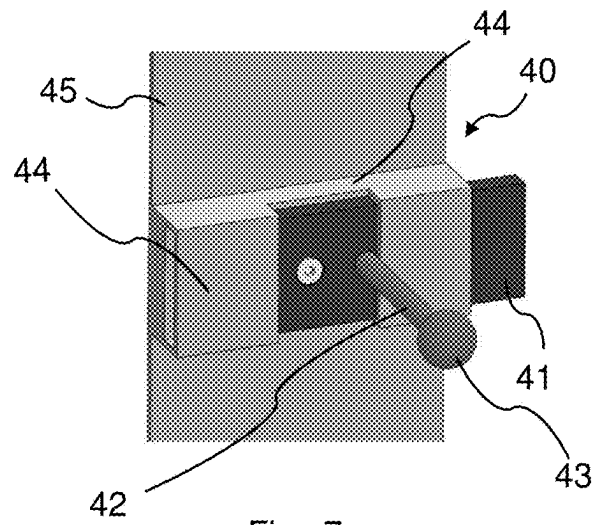


Fig. 7

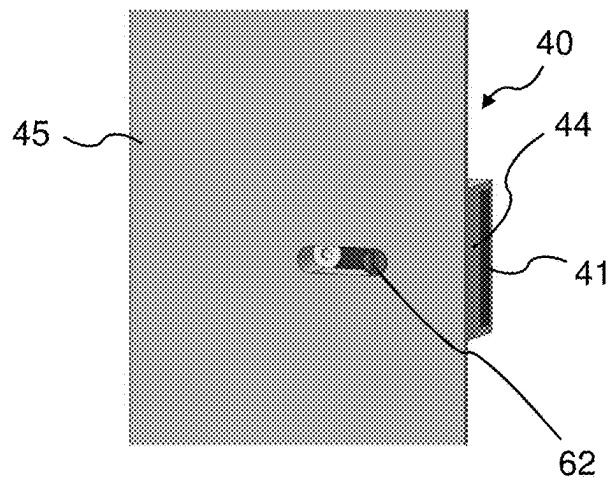


Fig. 8

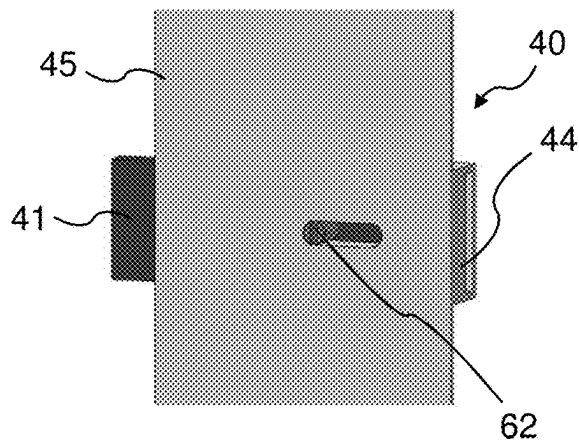


Fig. 9

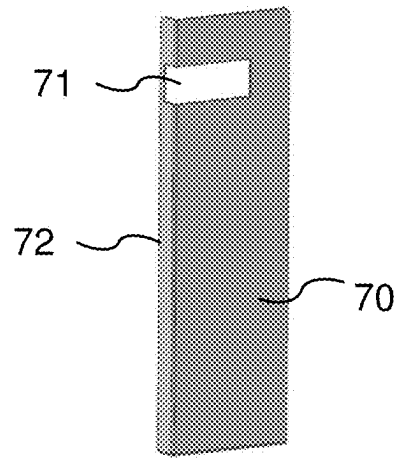


Fig. 10

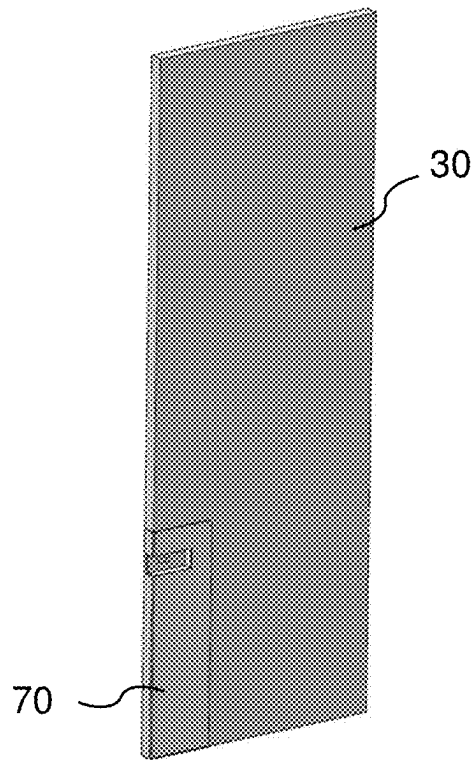
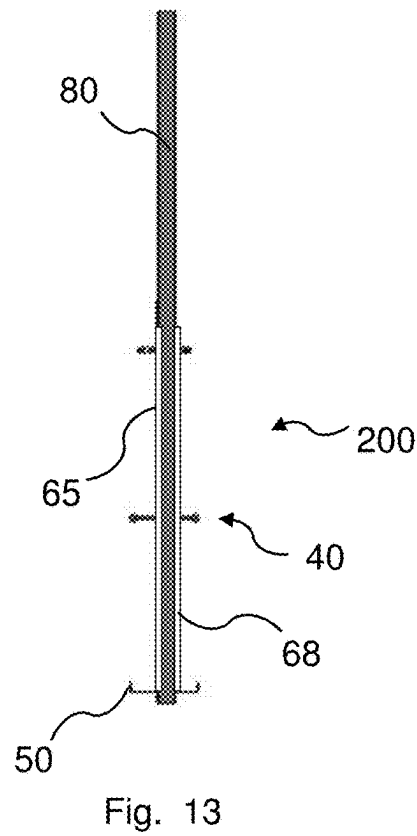
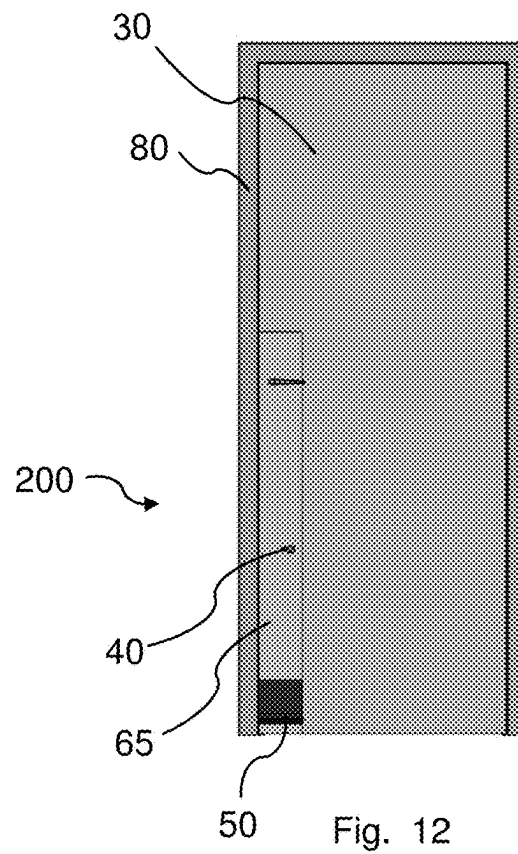


Fig. 11



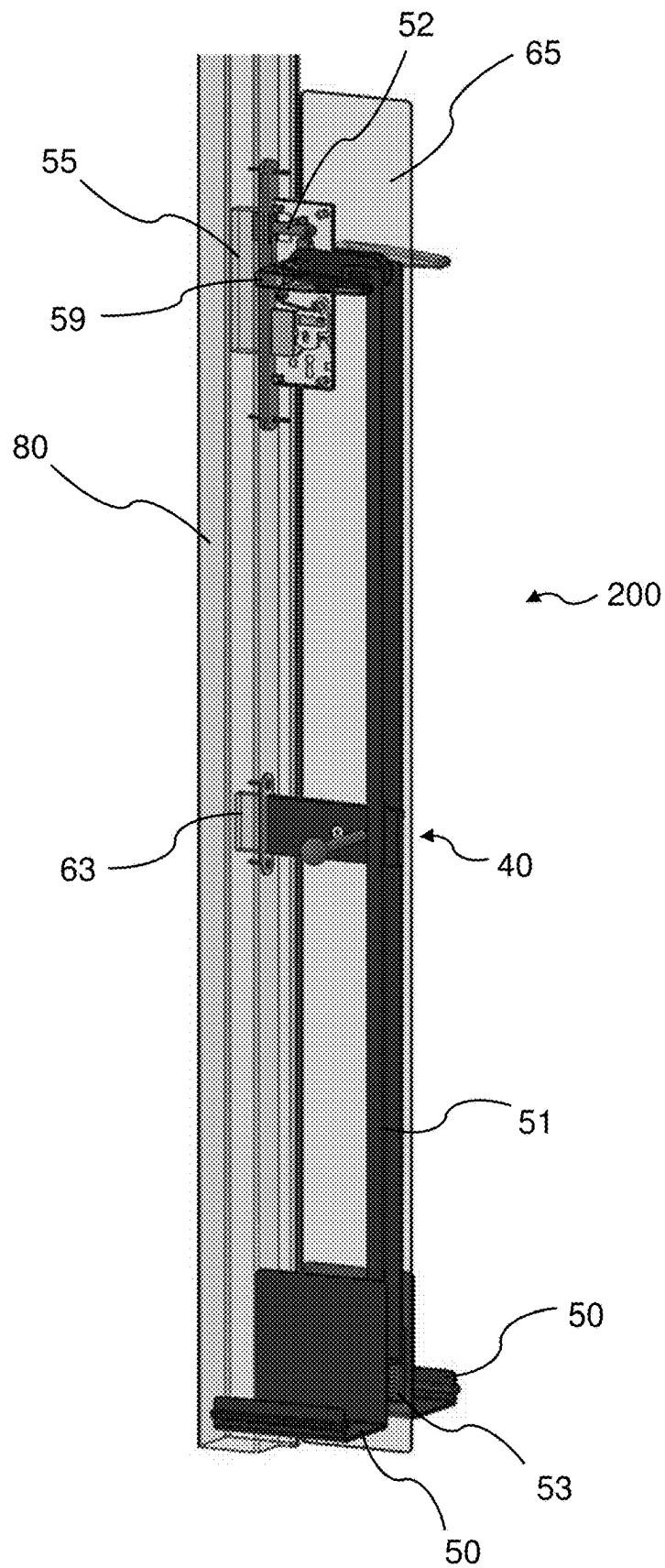


Fig. 14

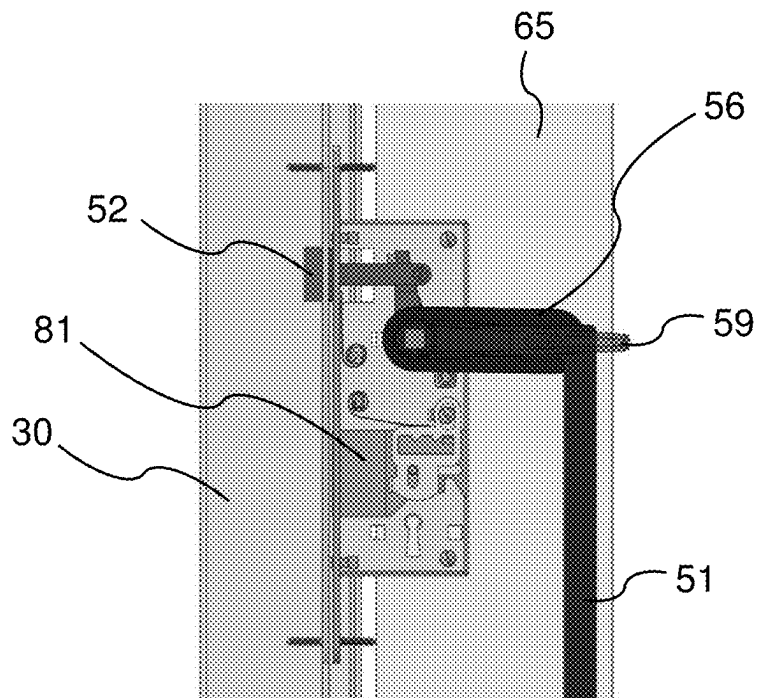


Fig. 15

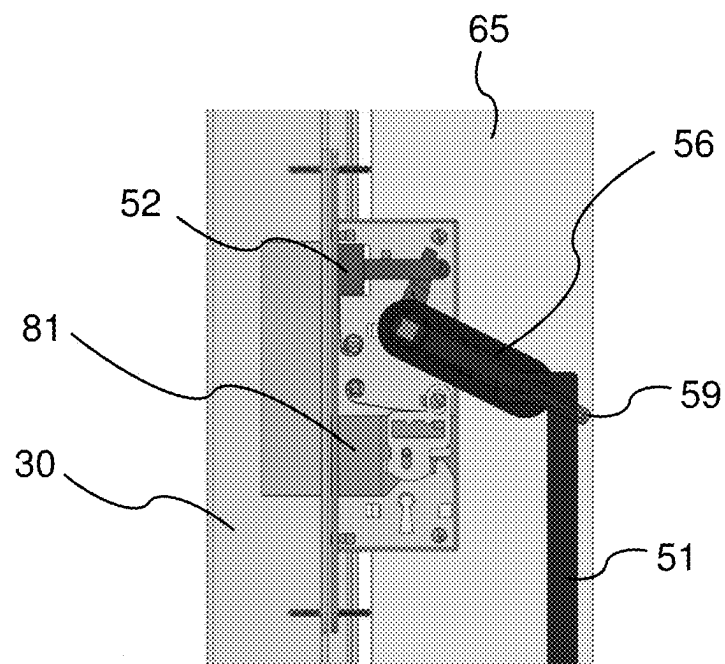


Fig. 16

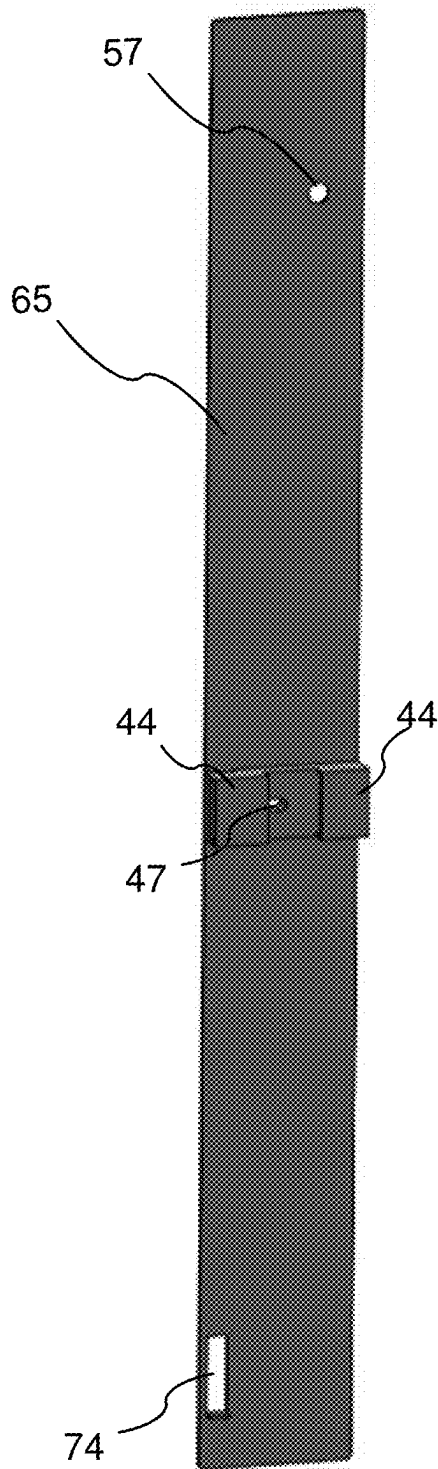


Fig. 17

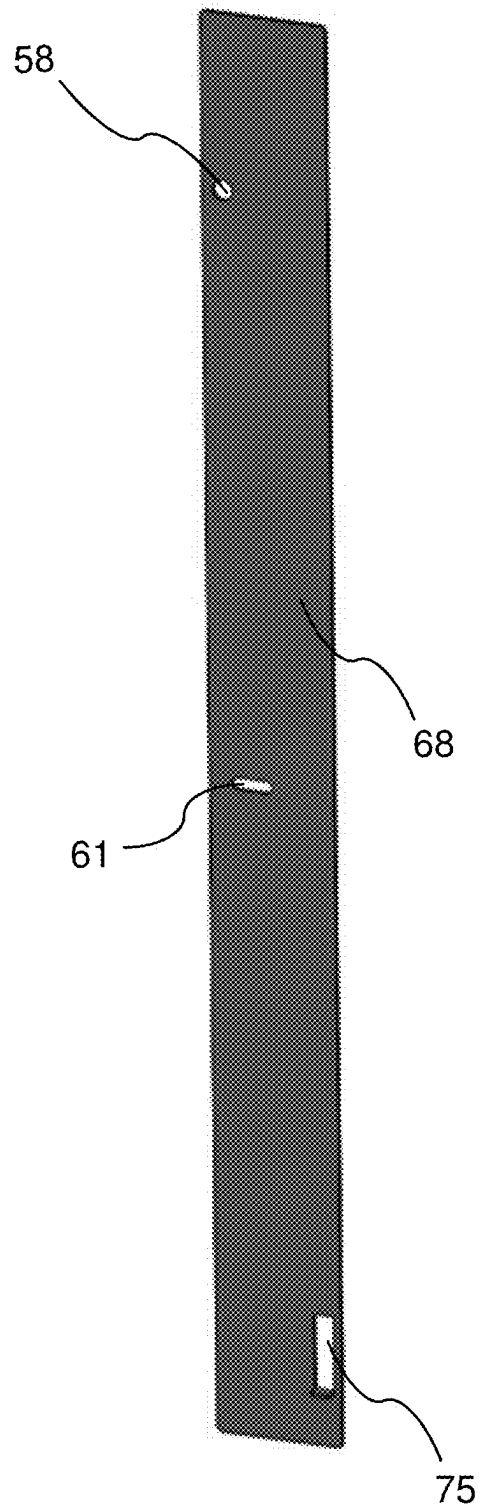


Fig. 18

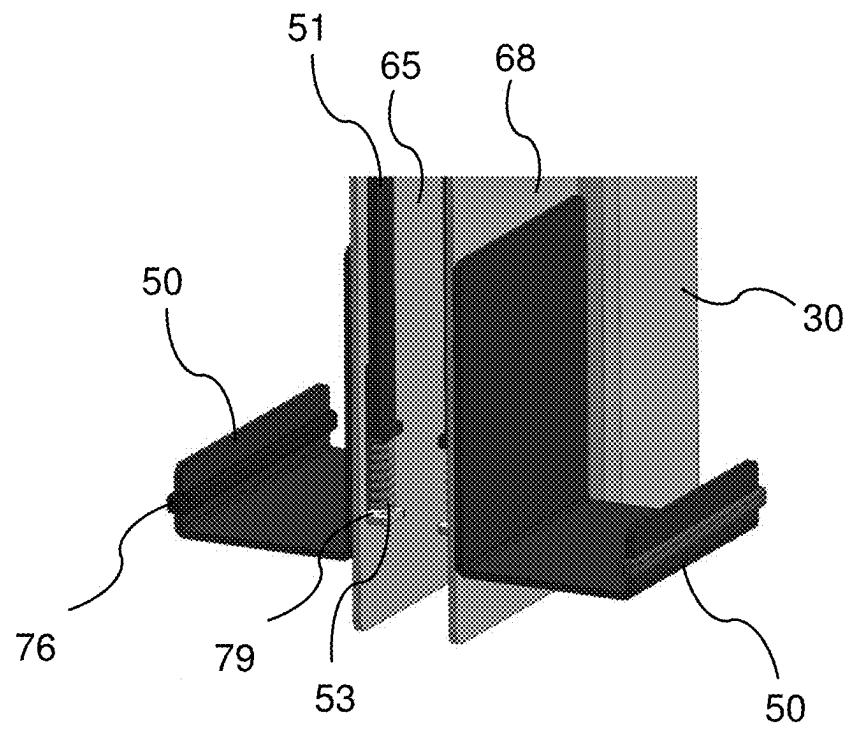


Fig. 19

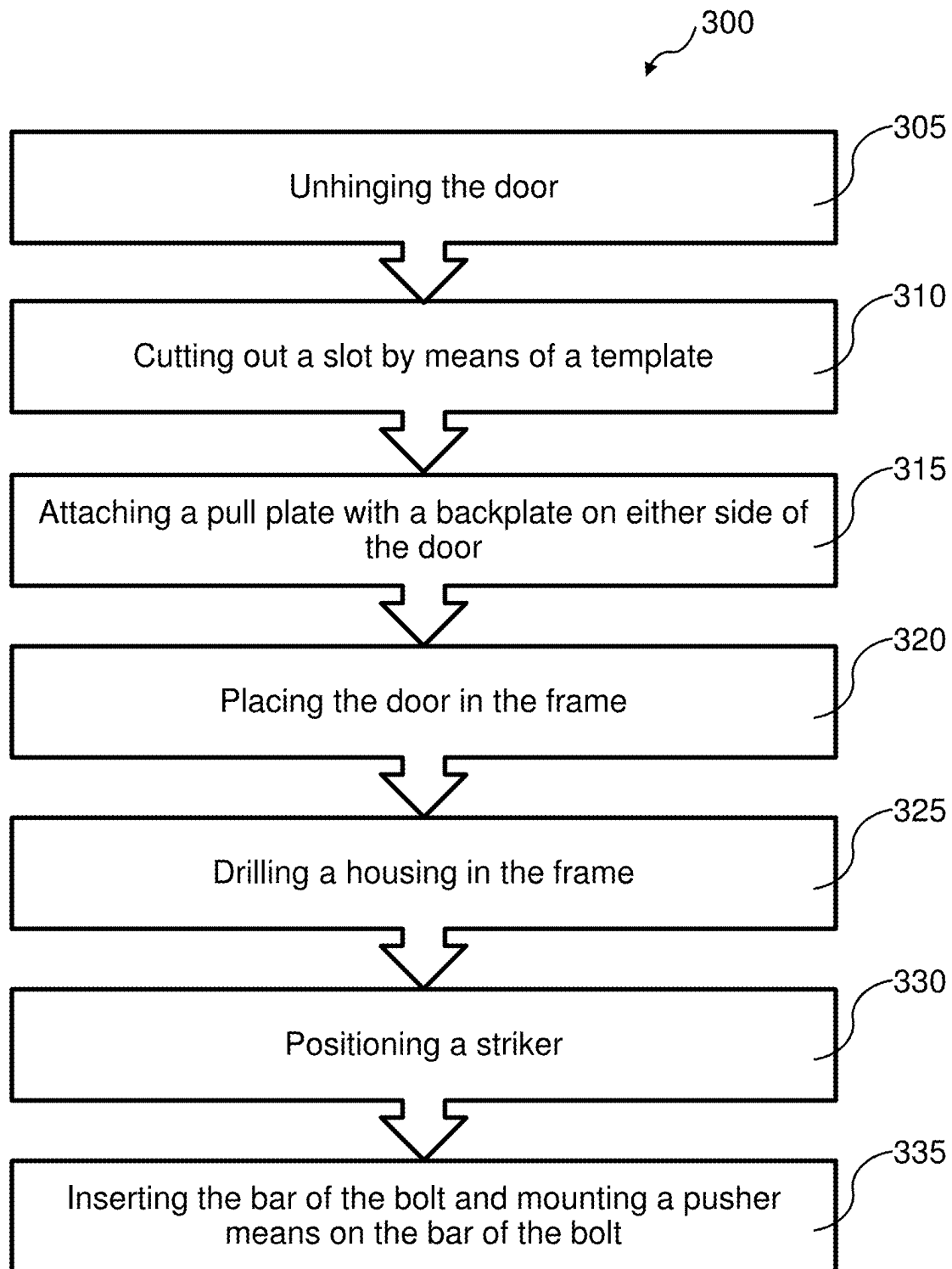


Fig. 20

DEVICE FOR LOCKING A DOOR**TECHNICAL FIELD OF THE INVENTION**

The present invention relates to a device for actuating and locking a door. The implementation of the invention is especially useful for the opening, closing and locking of a swing door with no contact of the user's hand with the door or with the actuation and locking device.

STATE OF THE ART

To limit the transmission of germs through hands coming into contact with the handles or surfaces of doors, it is desirable to keep these contacts to a minimum.

The prior art knows automatic doors whose opening is actuated by a motion sensor or weight sensor when detecting the presence of a user. However, the installation of this type of automatic door is costly and requires heavy maintenance.

The prior art knows doors mobile around a hinge with an arc of about 180 degrees and which the user can push with his foot or shoulder. This type of door has the drawback of requiring a large space for its installation, because the door can only be pushed.

In addition there is no practical system of use and implementation making it possible to lock a door with no contact between the user's hands and the handle or other surfaces of the door.

DESCRIPTION OF THE INVENTION

The present invention aims to remedy all or part of these drawbacks.

To this end, according to a first aspect, the invention envisages a device for locking a door, which comprises:

- a bolt comprising a bar configured to engage in a housing secured to the frame accommodating the door; and
- a pusher means secured to the bar;

the bolt being positioned on the door at a height of between thirty and eighty centimetres relative to the foot of the door, such that the bar of the bolt can be engaged with or disengaged from the housing secured to the frame by pressure applied by a user's knee against the pusher means; wherein the bar of the bolt slides in a panel secured to a pull plate fixed to the door; and wherein the pull plate comprises an extension positioned close to the foot of the door and configured to be driven by the foot or the lower leg of the user.

Thanks to these provisions, the door can be locked or unlocked by a user without needing to grasp a lock or a handle or to touch the surface of a door with the hand. Avoiding direct contact between the user's hand and the door or handle makes it possible to prevent infection through contact with a bacterium, virus or any other germ. As hands are known to be a main vector of infection because of their very frequent contact with the face and with all kinds of objects, the locking device improves hygiene and reduces the risk of infection between users of the same door.

The locking device that is the subject of the invention is suitable for implementation on any type of door. However, it is especially suitable for implementation on a frequently used door, for example in company premises or in public places, and on doors in places that are known sources of contamination, such as toilet doors.

Thanks to the bar of the bolt, which slides in a panel secured to a pull plate fixed to the door, and to the extension positioned close to the foot of the door and configured to be

driven by the foot or the lower leg of the user, the door can be driven by the user with no contact between his hands and the door. In addition, the pull plate prolongs the life of the device by spreading the forces, exerted over the surface of the pull plate by the user pushing or pulling on the device.

In some embodiments, the door is sandwiched between the pull plate and a backplate connected to the pull plate by fastening means.

Thanks to these provisions, the assembly formed of the pull plate, backplate and door is made more solid and long-lasting.

In some embodiments, the device comprises a through hole aligned with a grasping relief positioned on the bar, such that the bolt can be engaged or disengaged by a pressure exerted by means of a tool on the grasping relief, from the side of the door opposite the side on which the pusher means of the bolt is positioned.

Thanks to these provisions, the door can be locked or unlocked from the side of the door opposite the side on which the pusher means of the bolt is positioned. These provisions are especially useful for unlocking a door in an emergency or for locking a door when it is desirable to prohibit access.

In some embodiments, the free extremity of the pusher means is spaced at least four centimetres from the door by a stem. Preferably, the free extremity of the pusher means is spaced at an interval of at least six centimetres, for example eight centimetres. Thanks to these provisions, it is easier for the user's knee to exert pressure on the pusher means.

In some embodiments, the device comprises a pedal secured to a rod actuating a latch, such that the latch can be disengaged from a striker by pressure of a user's foot on the pedal.

Thanks to these provisions, the use of a handle to open the door is no longer necessary, and the disengagement of the latch and opening of the door can be achieved solely by the user's foot on the pedal.

In some embodiments, the pedal comprises an elastic return means for the pedal enabling the latch to be returned to a rest position.

Thanks to these provisions, in a similar way to the normal operation of a door handle, the latch is reengaged in the striker of the door when the striker and latch are aligned to each other and the user releases the pedal.

In some embodiments, the pedal is connected to the latch by a cable.

Thanks to these provisions, the device is particularly light and the links between a pedal and the latch can be incorporated into the body of the door.

In some embodiments, the link between the pedal and the latch is configured such that a movement of the latch does not cause the pedal to move.

Thanks to these provisions, the opening of the door by a manual action on the handle is not likely to cause the door to be blocked if there is an object under the pedal.

In some embodiments, the pedal comprises a damper flange.

This flange makes it possible to prevent the wall being marked when the pedal touches it.

In some embodiments, the extension comprises a damper flange.

This flange makes it possible to prevent the wall being marked when the extension touches it.

In some embodiments, the surface of the door intended to be pushed is equipped with a backplate, thereby allowing the door to be opened by pushing with the foot.

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According to a second aspect, the present invention envisages a door comprising a device that is the subject of the invention.

In some embodiments, at least one pull plate is incorporated into the body of the door and cannot be seen from the outside of the door, except for its extension, when the door is closed.

According to a third aspect, the present invention envisages a method for installing a locking device for a door that is the subject of the invention, which comprises the following steps:

- unhinging the door from its frame;
- cutting out a slot by means of a template equipped with a window having dimensions matching the dimensions of the panel of a bolt, at a height relative to the foot of the door of between thirty and eighty centimetres;
- positioning the panel of the bolt in the slot and attaching a pull plate with a backplate on either side of the door;
- positioning an extension of the pull plate close to the foot of the door, an extension configured to be driven by the foot or the lower leg of the user;
- placing the door in the frame;
- drilling a housing in the frame, positioning a striker or mounting a stud, at the height of the bolt; and
- inserting the bar of the bolt and mounting a pusher means on the bar of the bolt, the bar of the bolt sliding in the panel secured to the pull plate.

Thanks to these provisions, the locking device according to the invention can be installed on any type of door already in place without difficulty. These provisions also have the advantage of enabling an existing set of doors to be adapted, rather than replaced, in order to address a health requirement.

As the particular aims, advantages and features of the door that is the subject of the invention and the method that is the subject of the invention are similar to those of the device that is the subject of the invention, they are not repeated here.

BRIEF DESCRIPTION OF THE FIGURES

Other advantages, aims and particular features of the invention will become apparent from the non-limiting description that follows of at least one particular embodiment of the device and method that are the subjects of the present invention, with reference to drawings included in an appendix, wherein:

FIG. 1 represents, schematically and in a front view, a first particular embodiment of the locking device that is the subject of this invention;

FIG. 2 represents, schematically and in a side view, the locking device shown in FIG. 1;

FIG. 3 represents, schematically and in a perspective view, a detail of the bolt, in the open position, of the locking device that is the subject of this invention shown in FIGS. 1 and 2;

FIG. 4 represents, schematically and in a perspective view, the detail of the bolt shown in FIG. 3, in the closed position;

FIG. 5 represents, schematically and in a perspective view, two views side by side of a single backplate, implemented in the locking device shown in FIGS. 1 to 4, according to two different viewing angles;

FIG. 6 represents, schematically and in a perspective view, two views side by side of a single pull plate, implemented in the locking device shown in FIGS. 1 to 5, according to two different viewing angles;

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FIG. 7 represents, schematically and in a perspective view, a detail of a bolt of the locking device shown in FIGS. 1 to 6, in the closed position;

FIG. 8 represents, schematically and in a perspective view, the detail of a bolt of the locking device shown in FIGS. 1 to 7, in the open position;

FIG. 9 represents, schematically and in a perspective view, the detail of a bolt implemented in the locking device shown in FIGS. 1 to 8, in the closed position;

FIG. 10 represents, schematically and in a perspective view, a template used in a particular embodiment of the method for installing a locking device that is the subject of the invention;

FIG. 11 represents, schematically and in a perspective view, the template of FIG. 10, placed against a door during the implementation of the method for installing a locking device that is the subject of the invention;

FIG. 12 represents, schematically and in a front view, a second particular embodiment of the locking device that is the subject of the invention;

FIG. 13 represents, schematically and in a side view, the locking device shown in FIG. 12;

FIG. 14 represents, schematically and in a perspective view, the locking device shown in FIGS. 12 and 13;

FIG. 15 represents, schematically, the actuating mechanism of the latch implemented in the locking device shown in FIGS. 12 to 14, in the closed position;

FIG. 16 represents, schematically, the actuating mechanism of the latch implemented in the locking device shown in FIGS. 12 to 15, in the open position;

FIG. 17 represents, schematically and in a perspective view, a pull plate, implemented in the locking device shown in FIGS. 12 to 16;

FIG. 18 represents, schematically and in a perspective view, a backplate, implemented in the locking device shown in FIGS. 12 to 17;

FIG. 19 represents, schematically and in a perspective view, a detail of the pedal of the locking device shown in FIG. 14, according to another viewing angle; and

FIG. 20 represents, schematically and in the form of a logic diagram, a series of steps of a particular embodiment of the method for installing a locking device for a door, that is the subject of the invention.

DESCRIPTION OF THE EMBODIMENTS

The present description is given in a non-limiting way, in which each characteristic of an embodiment can be combined with any other characteristic of any other embodiment in an advantageous way.

Note that the figures are to scale, but the various figures may be to different scales.

FIGS. 1 and 2 show a schematic view of a first particular embodiment of the locking device 100 that is the subject of the invention.

The locking device 100 makes it possible to lock a door 30 accommodated in a frame 80, more commonly referred to as a doorframe. The device 100 comprises a bolt 40 positioned at a height of between thirty and eighty centimetres from the floor, i.e. from the lowest edge of the door 30, preferably between forty and eighty centimetres from the floor, and more preferably between forty and seventy centimetres from the floor. Preferably the bolt 40 is positioned at a height of between forty-five and fifty-five centimetres, for example at fifty-five centimetres from the floor.

The bolt 40 is described with reference to FIGS. 3 and 4, and 7 to 9.

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The device **100** comprises a pull plate **45** comprising a flat surface affixed on a surface of the door **30**, and a backplate **48** comprising a flat surface affixed on the other surface of the door **30**. The pull plate **45** and the backplate **48** are preferably attached to each other by fastening means, for example by clamping bolts (not shown) going through the door **30** so as to enclose the door **30**.

The characteristics of the plate **45** and backplate **48** are described with reference to FIGS. **5** and **6**.

FIGS. **3** and **4** show the bolt **40** mounted on a pull plate **45**. The bolt **40** comprises a bar **41** mounted free in translation in a panel **44**. Preferably, the panel **44** is monolithic with the pull plate **45** and comprises two lugs that support and constrain the movement of the bar **41**. It is specified that the two lugs shown in FIGS. **3** and **4** are represented partially transparent.

The translational movement of the bar **41** in the panel **44** is constrained by two stops. The bolt **40** can adopt two positions corresponding to the two extreme positions of the bar **41**.

In FIG. **3**, the bolt **40** is in the open position, i.e. completely disengaged from the housing **63** made in the frame **80**. Preferably, the housing **63** comprises a striker. In this open position of the bolt **40**, the door **30** can be opened. In FIG. **4**, the bolt **40** is in the closed position, i.e. engaged in the housing **63**. In this closed position of the bolt **40**, the door **30** is locked in place.

The translation of the bar **41** in the panel **44** is actuated by the application of a lateral force on a pusher means **43** secured to the bar **41**. Preferably, a stem **42** connects the bar **41** to the pusher means **43** so that the pusher means is clear of the surface of the door to facilitate its actuation by the knee. Preferably, the stem **42** passes through an oblong hole **61** drilled in the backplate **48**. The two extremities of the oblong hole **61** are, in this case, the stops limiting the translational movement of the bar **41**.

The bolt **40** is positioned on the door **30** at a height relative to the foot of the door of between thirty and eighty centimetres, preferably between forty and eighty centimetres, and more preferably between forty and seventy centimetres. The bar of the bolt can be engaged with or disengaged from the housing secured to the frame **80** by pressure of a user's knee against the pusher means **43**.

In addition, the free extremity of the pusher means **43** is spaced at least four centimetres from the door **30** by the stem **42**. Preferably, the free extremity of the pusher means **43** is spaced at least six centimetres from the door **30** by the stem **42**, e.g. six centimetres.

FIG. **5** shows a single backplate **48** according to two different viewing angles. FIG. **6** shows a single pull plate **45** according to two different viewing angles.

The pull plate **45** comprises the panel **44** of the bolt **40**. The pull plate **45** is pierced by a hole **47** whose function is described with reference to FIGS. **7** to **9**.

The backplate **48** comprises an oblong through hole **61** dimensioned for the stem **42** of the bolt **40** to pass through.

The pull plate **45** and the backplate **48** each comprise an extension, respectively **46** and **49**, positioned close to the foot of the door and configured to be driven by the foot or the lower leg of the user. For example, the extension, **46** or **49**, has a "U" shape or a hook shape that can be grasped by the foot or by the heel of the user.

In some embodiments, the extension, **46** or **49**, comprises a damper flange. For example, the damper flange is a ring made of artificial rubber positioned on the perimeter of the extension, **46** or **49**.

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FIGS. **7**, **8** and **9** represent, schematically and in a perspective view, the detail of the bolt **40** implemented in the first embodiment of the device **100**. FIGS. **7** and **9** show the bolt **40** in a closed position. FIG. **8** shows the bolt **40** in an open position.

The pull plate **45** is pierced by a through hole **61**, aligned with a grasping relief **62** positioned on the bar **41** or secured to it. The through hole **61** allows the passage of a tool, for example a rigid stick, a key or a screwdriver, such that the tool can be pressed against the grasping relief **62** to engage or disengage the bolt **40**.

FIGS. **10** and **11** show a template **70** useful for implementing the method for installing a locking device that is the subject of the invention, a method detailed with reference to FIG. **20**. The template **70** comprises at least one band **72** bent at a right angle and intended to be abutted against a section of the door **30** in its largest dimension, i.e. its height. Preferably, the template **70** comprises a second band bent at a right angle, intended to abut against the section of the door **30** in its intermediate dimension, i.e. its width, as shown in FIG. **11**. Therefore, the template **70** has a shape complementary with the corner of a door **30** and serves as a model for cutting out a cavity having dimensions matching the dimensions of the panel **44** of the bolt **40**. FIGS. **12** and **13** represents, schematically, a second particular embodiment of the locking device **200** that is the subject of the invention.

The locking device **200** makes it possible to lock a door **30** accommodated in a frame **80**. The device **200** comprises a bolt **40** positioned at a height relative to the foot of the door of between thirty and eighty centimetres, preferably between forty and eighty centimetres, and more preferably between forty and seventy centimetres. Preferably, the bolt **40** is similar to the one already described above.

The device **200** comprises a pull plate **65** comprising a flat surface affixed on a surface of the door **30**, and a backplate **68** comprising a flat surface affixed on the other surface of the door **30**. The pull plate **65** and the backplate **68** are preferably attached to each other by fastening means, for example by clamping bolts (not shown) going through the door **30** so as to enclose the door **30**.

The characteristics of the pull plate **65** and backplate **68** are described with reference to FIGS. **17** and **18**.

FIG. **14** represents, schematically and in perspective, the locking device **200**. It is stated that the door **30** and backplate **68** are not represented here in order to better show the other elements of the locking device **200**.

The device **200** differs from the device **100** described above in that it comprises a pedal **50** secured to a rod **51** actuating a latch **52**, such that the latch **52** can be disengaged from a striker **55**, mounted in the frame **80**, by pressure of a user's foot on the pedal **50**. This mechanism is accommodated in a space delimited by the pull plate **65** and the backplate **68** and is therefore not visible under the normal condition of use of the locking device **200**.

In some embodiments, the pull plate **65** and backplate **68** are elongated to cover the entire height of the door **30**. Preferably, the pull plate **65** and backplate **68** are made from a material identical to that of the handle **59**, when this is retained, to harmonise the whole. The pull plate **65** and backplate **68** can comprise any type of material, in particular stainless steel, aluminium, alloys and/or plastic polymers, and/or composite materials. Preferably, the pull plate **65** and backplate **68** comprise stainless steel.

In some embodiments, a door number or a pictograph is imprinted on the pull plate **65** or backplate **68** or a panel is provided for this use.

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In some embodiments, the device **200** comprises an additional part making it possible to keep the door open when it is equipped with a door closer. It consists, for example, of a wedge, a hook or a door chain.

In some embodiments, a vertical slot, which extends from the pedal to the bolt and has dimensions matching the dimensions of the rod **51**, is formed in the pull plate **65** and/or in the backplate **68**. Therefore, the rod **51** can be positioned in the slot.

FIGS. **15** and **16** show the actuating mechanism of the latch **52**, implemented in the device **200**. This mechanism comprises the latch **52** connected to the rod **51** by means of a pivot **56** configured to convert a vertical traction force exerted on the stick into a horizontal traction force on the latch **52**. Any other similar type of mechanical link well known to the person skilled in the art can serve here as a pivot.

In FIG. **15**, the actuating mechanism of the latch is represented in a closed position, i.e. with the latch **52** engaged in the striker **55** mounted in the frame **80**. Conversely, in FIG. **16**, the actuating mechanism of the latch is represented in an open position. In some embodiments, the pivot **56** is also secured to a conventional handle **59**, thereby enabling the latch **52** to be disengaged either by means of the pedal or by means of the handle **59**, at the user's choice.

FIG. **19** shows, schematically and in a perspective view, the lower portion of the device **200** shown in FIG. **14**, according to another viewing angle. An assembly formed of the rod **51** and the pedal **50** is mounted on the pull plate **65**. The force exerted by the user's foot on the pedal **50** actuates the downward movement of the rod **51**. The pedal **50** comprises a damper flange **76**. Therefore, the door stop fixed in the ground of the prior art can be absent, which makes maintenance of the floor easier.

In some embodiments, the pedal **50** comprises an elastic return means for the pedal enabling the latch **52** to be returned to a rest position, actuated by the movement of the pedal. Preferably, the return means is a spring **53** pressed against a stop **79** by the rod **51** when a force is exerted on the pedal **50** secured to the rod **51**.

In some embodiments, the same pedal system is mounted on the backplate **68**.

In some embodiments, the extension, **46** or **49**, comprises a damper flange **76**. For example, the damper flange is a ring made of artificial rubber positioned on a curved portion at the free extremity of the pedal **50**.

FIG. **17** represents, schematically and in a perspective view, the pull plate **65**, and FIG. **18** represents, schematically and in a perspective view, the backplate **68** of the locking device **200**. The elements described above, in particular in FIGS. **5** and **6**, are not described again here. The pull plate **65** and backplate **68** differ from the corresponding plates of the first embodiment **100** in that they are extended up to the height of the handle **59**, at about one metre from the floor. The pull plate **65** and backplate **68** also differ from the corresponding plates of the first embodiment **100** in that they comprise an opening, **74** and **75**, dimensioned to accommodate a spring **53** and a mechanical link between the pedal **50** and the rod **51**. In addition, the pull plate **65** and backplate **68** are pierced with holes, **57** and **58**, making it possible to accommodate the axis of a door handle **59** making it possible to actuate the pivot shown in FIGS. **15** and **16**.

The link between the pedal **50** and the latch **52** is preferably configured such that a movement of the latch **52** does not cause the pedal **50** to move. By providing for a stop on the sliding link of the pedal **50** on the rod **51**, the descent of the pedal **50** causes the rod **51** to descend, but the decent of the

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rod **51** does not cause the pedal **50** to descend. This therefore avoids a risk of the door jamming if the handle **59** is operated by hand, especially if there is an object under the pedal **50**. Of course, a cable link operating in tension but not in compression (e.g. a cable that is not guided in a sleeve) makes the same technical result possible.

In some embodiments (not shown), the device that is the subject of the invention is incorporated in the volume of the door, only the means for actuation by the knee and/or foot protruding from the surface of the door. In these embodiments, the pull plates **45** and **65** are incorporated in the body of the door and are therefore, except for their extension **46** or **49**, not visible from the outside of the door when the door is closed, and possibly also when the door is open.

In some embodiments (not shown), the link between the pusher means **43** that can be actuated by the knee and the bar **41** of the bolt **40** is a cable link, not a link using at least one rigid part.

In some embodiments (not shown), the link between the pedal **50** and the latch **52** allowing the latch **52** to be disengaged from the striker **55** by pressure of a user's foot on the pedal **50** is a cable link, not a link using rigid parts. Preferably, the cable is not completely within a sleeve for operating in tension but not in compression. This therefore avoids a downward movement of the handle, for example through the action of a user's hand, leading to a downward movement of the pedal, which may result in the door being blocked, especially if there is an object under the pedal.

In some embodiments, the actuation of the bolt **40** causes the actuation of the latch **52**. For example, a cable guided by pulleys exerts traction on the deadbolt **81** when the bar **41** exits from the housing **63**. Of course, a linkage with a stop can replace a cable guided by pulleys to obtain the same mechanical effect.

In some embodiments, the latch is replaced by an olive and/or the lock is of the mortice lock type.

FIG. **20** represents, schematically and in the form of a logic diagram, a series of steps of a particular embodiment of the method **300** for installing a locking device for a door, that is the subject of the invention.

The method **300** for installing a locking device for a door comprises the following steps:

305, unhinging the door from its frame;

310, cutting out a slot by means of a template equipped with a window having dimensions matching the dimensions of the panel of a bolt, at a height relative to the foot of the door of between thirty and eighty centimetres and, preferably, between forty and eighty centimetres, and more preferably between forty and seventy centimetres;

315, positioning the panel of the bolt in the slot and attaching a pull plate with a backplate on either side of the door;

320, placing the door in the frame;

drilling a housing in the frame, **330** positioning a striker or mounting a stud, at the height of the bolt; and

335, inserting the bar of the bolt and mounting a pusher means on the bar of the bolt.

In particular, the step **310** of cutting out a slot by means of a template can be carried out by means of the template shown in FIGS. **10** and **11**.

Preferably, the bolt panel is integrally formed in the pull plate. These provisions reduce the number of parts of the device and facilitate its placement. In addition, the positioning of the panel of the bolt in the slot makes it possible, in this case, to easily align the pull plate in the appropriate position.

In some embodiments, the method 300 for installing a locking device for a door also comprises the following steps (not shown):

drilling holes configured to accommodate screws connecting a pull plate, equipped with the panel of the bolt, and a backplate;

placing a pivot and a rod, configured to be secured to a pedal;

placing a return means (spring) between the rod and the pedal, and placing the pedal; and

placing the door in the frame.

The invention claimed is:

1. A device for locking a door, comprising:

a bolt comprising a bar configured to engage in a housing secured to the frame accommodating the door; and

a pusher means secured to the bar;

the bolt being positioned on the door at a height of between thirty and eighty centimetres relative to a foot of the door, such that the bar of the bolt is configured to slide horizontally in a left or right sliding direction so that the bar is engaged with or disengaged from the housing secured to the frame by a pressure, applied in the sliding direction, applied by a user's knee against the pusher means;

wherein the bar of the bolt slides in a panel secured to a pull plate fixed to the door, and

wherein the pull plate comprises an extension positioned close to the foot of the door and configured to be driven by a foot or a lower leg of the user, the extension and the pull plate forming a hook shape.

2. The device according to claim 1, wherein the door is sandwiched between the pull plate and a backplate connected to the pull plate by fastening means.

3. The device according to claim 1, which comprises a through hole aligned with a grasping relief positioned on the bar, such that the bolt is configured to be slidably engaged or disengaged by a lateral pressure exerted by means of a tool on the grasping relief, from the side of the door opposite the side on which the pusher means of the bolt is positioned.

4. The device according to claim 1, wherein the free extremity of the pusher means is spaced at least four centimetres from the door by a stem.

5. The device according to claim 1, which comprises a pedal secured to a rod actuating a latch, such that the latch is configured to be disengaged from a striker by pressure of a user's foot on the pedal.

6. The device according to claim 5, wherein the pedal comprises an elastic return means for the pedal enabling the latch to be returned to a rest position.

7. The device according to claim 5, wherein the pedal is connected to the latch by a cable.

8. The device according to claim 5, wherein the link between the pedal and the latch is configured such that a movement of the latch does not cause the pedal to move.

9. The device according to claim 5, wherein the pedal comprises a damper flange.

10. The device according to claim 1, wherein the extension comprises a damper flange.

11. The device according to claim 1, wherein a surface of the door intended to be pushed is equipped with a backplate, thereby allowing the door to be opened by pushing with the foot.

12. A door comprising a device according to claim 1.

13. The door according to claim 12, wherein at least one pull plate is incorporated into the body of the door and cannot be seen from the outside of the door, except for its extension, when the door is closed.

14. A method for installing a device for locking a door according to claim 1, characterized in that it comprises the steps of:

unhinging the door from its frame;

cutting out a slot corresponding to the shape of a template equipped with a window having dimensions matching the dimensions of the panel of the bolt, at a height relative to the foot of the door of between thirty and eighty centimetres;

positioning the panel of the bolt in the slot and attaching a pull plate with a backplate on either side of the door;

positioning the extension of the pull plate close to the foot of the door, an extension configured to be driven by the foot or the lower leg of the user;

placing the door in the frame;

drilling a housing in the frame, positioning a striker or mounting a stud, at the height of the panel of the bolt; and

inserting the bar of the bolt and mounting the pusher means on the bar of the bolt, the bar of the bolt sliding in the panel secured to the pull plate.

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